

Experiment 2: Fixed Height + Fixed Wind + Obstacle Ablation | no_obstacle | Comparison Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1991	2.469	1.342	5.689	36.1%	69.1%	81.7%	14.119	28.574	9.110	23.049	0.569	1.304	1.441
PPO-GRU	1989	3.050	0.819	13.533	61.1%	81.3%	83.4%	7.824	18.474	6.237	13.751	0.367	0.851	1.117
PPO-RNN	1993	2.498	1.227	5.975	40.0%	73.2%	84.7%	16.881	38.824	7.315	15.199	0.491	1.315	1.470
MPC	1071	6.092	3.094	16.748	2.9%	17.3%	45.4%	2.839	3.682	1.475	2.638	0.143	0.145	0.172

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1991	99.5%	9	0.5%	0.5%	9	100.0%	0	0
PPO-GRU	2000	1989	99.5%	11	0.6%	0.6%	11	100.0%	0	0
PPO-RNN	2000	1993	99.7%	7	0.3%	0.3%	7	100.0%	0	0
MPC	2000	1071	53.5%	929	46.5%	46.5%	929	100.0%	0	0

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	929	922	99.2%	227	24.6%	2.856
PPO-GRU	929	920	99.0%	148	16.1%	3.077
PPO-RNN	929	924	99.5%	140	15.2%	2.536